

(21) Application No: 1909336.8	(51) INT CL: D06M 11/74 (2006.01) A41D 1/00 (2018.01) A61B 5/00 (2006.01)
(22) Date of Filing: 28.06.2019	(56) Documents Cited: WO 2017/009879 A1 CN 108403097 A CN 107951473 A US 20190132948 A1 Xu, Z. and Gao, C. "Graphene fiber: a new trend in carbon fibers", Materials today, November 2015, volume 18, number 9, pages 480-492, Elsevier Guan-Hang Y. et.al. "Graphene fibers: advancing applications in sensor, energy storage and conversion", Chinese journal of polymer science, 2019, volume 37, pages 535-547
(71) Applicant(s): Prevayl Limited 10 West Street, Alderley Edge, Cheshire, SK9 7EG, United Kingdom	(58) Field of Search: INT CL A61B, D06M Other: WPI, EPODOC, INTERNET, XPESP, XSPRNG
(72) Inventor(s): Tahir Mahmood	
(74) Agent and/or Address for Service: Prevayl Limited 10 West Street, Alderley Edge, Cheshire, SK9 7EG, United Kingdom	

(54) Title of the Invention: **Biosensing textile and garment**
Abstract Title: **Biosensing textile used in clothing**

(57) The textile includes a distributed sensing system. The sensing system comprises a controller in communication with and controlling sensor electrodes on the textile. The sensor electrodes are formed of two-dimensional (2D) electrically conductive material, preferably graphene. Graphene forming the electrodes and electrical conductors is preferably printed or transferred onto the cloth. Alternatively, the conductor comprises conductive fibres or yarns forming the fabric. The preferred system includes a rechargeable battery, energy harvesting device, communicator operable over a 4G or 5G wireless cellular network. The textile is formed into a garment, especially shirt, t-shirt, blouse, dress, brassiere, shorts, trousers, vest, jacket, coat, glove, armband, underwear, headband, hat, cap, collar, waistband, stocking, sock, show, swimwear, wetsuit, drysuit or athletic clothing.

